

Zhanggir Yessenaliyev

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EDUCATION

Purdue University

B.S. in Robotics Engineering Technology

Coursework: Object-Oriented Programming, C programming, MATLAB, Circuits & Soldering, Multivariable Calculus, Geometric Modeling & PDM

West Lafayette, IN

Aug 2025 – May 2029

TECHNICAL SKILLS

Languages: C++17, C, Python, MATLAB, Bash, HTML/CSS

AI/Computer Vision: PyTorch, CUDA, OpenCV, NumPy, Ultralytics YOLO, Image Segmentation

Robotics: ROS 2, CARLA, Nav2, Gazebo, SLAM Toolbox, TF2, Localization, SLAM, Sensor Fusion, LiDAR

Developer Tools: Linux, Git, Docker, CMake, RViz2, Foxglove

CAD/CAE: SOLIDWORKS, NX, Autodesk Fusion 360

EXPERIENCE

Purdue IEEE Autonomous Kart Racing

Controls & Perception Team Member

Jan 2026 – Present

West Lafayette, IN

- Built a 30 Hz ROS 2 localization stack fusing ZED VIO and IMU via an EKF with 100 ms sensor timeouts.
- Migrated odom to base.link TF ownership from the ZED node to robot_localization for a robot-centric tree.
- Configured ZED GEN3 tracking with gravity-aligned origin and 2D mode, enabling autonomous kart testing.
- Enabled deployment of the kart at the Autonomous Kart Series by validating its localization systems.

SilkRoad Innovation Hub – Early-Stage VC

Robotics Fellow

Jul 2026 – Aug 2026

Menlo Park, CA

- Earned selection as 1 of 10 Fellows from a competitive pool of 120+ applicants.
- Presented robotics projects to engineers from Uzbekistan’s Ministry of Education and advised them on investor readiness.

Engineering Club

Science Mentor

Oct 2024 – Jun 2025

Almaty, Kazakhstan

- Recruited and led a 15-student engineering team, assigning development responsibilities for a biomechanical hand.
- Designed, prototyped, and tested an EMG-controlled biomechanical hand using an Arduino-based control system.
- Placed 3rd among 656 projects at InfoMatrix Asia.

PROJECTS

Vision-Guided Autonomous Navigation | Python, PyTorch, CUDA, CARLA, OpenCV

Jul 2026 – Sep 2026

- Built an autopilot-free, route-aware controller that achieved 98.9% mean completion across 9 weather-varied runs over 1.48 km with 0 collisions and 0 solid-line crossings.
- Accelerated model inference 20.4x over CPU to 199 FPS with 4.72 ms median latency using CUDA mixed precision.
- Trained a 117K-parameter U-Net-style deep neural network in PyTorch on 900 synchronized CARLA RGB/semantic pairs, achieving 92.3% IoU and 96.0% Dice on 300 held-out images.

Occlusion-Aware Augmented Reality Localization | Python, OpenCV, NumPy, ArUco

Jul 2026 – Present

- Built a world-space camera-geometry pipeline implementing lens correction, coordinate transforms, and monocular ground-plane projection.
- Achieved 0.57-pixel calibration RMS across 20+ views and implemented ArUco/solvePnP camera localization with reprojection-error diagnostics.

Nav2 Custom Controller | C++, ROS 2 Jazzy, Nav2, pluginlib, TF2

Jun 2026 – Aug 2026

- Implemented a 20 Hz C++ Pure Pursuit path-tracking algorithm using Nav2 global paths and robot poses to generate bounded velocity commands in 0.2 ms per control cycle.
- Integrated pluginlib lifecycle management, TF2 transformations, costmap-aware path pruning, RViz visualization, and 10 YAML-configurable motion parameters.
- Automated TurtleBot3 testing across 20 navigation goals by scripting simulation launch, AMCL initialization, navigation-goal dispatch, and headless or graphical execution.

Autonomous Differential Rover Sim | ROS 2 Jazzy, Nav2, Gazebo, SLAM, URDF/Xacro, Python

Jul 2026

- Built a URDF/Xacro differential-drive rover with Gazebo physics, 50 Hz odometry, and 360-degree LiDAR.
- Bridged six Gazebo–ROS 2 channels for control, odometry, TF, joint states, LiDAR, and simulation time.
- Configured SLAM Toolbox for 5 cm occupancy-grid mapping with scan matching and loop closure.

Little Blazer Engine CAD Assembly | Siemens NX, Teamcenter, PDM/PLM

May 2026

- Built a complete Little Blazer Engine assembly in Siemens NX using parametric parts and subassemblies.
- Produced engineering drawings and 3D annotations documenting component geometry and assembly relationships.
- Managed revision-controlled parts and assembly data in Teamcenter using PDM/PLM workflows.